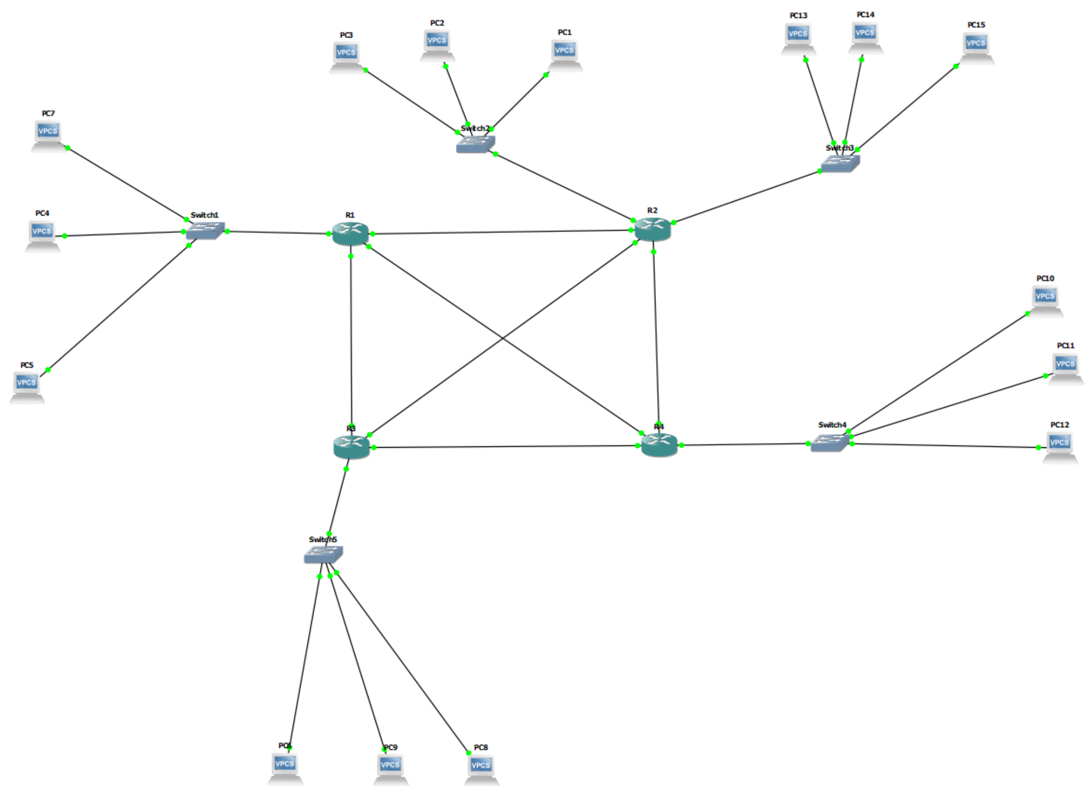
PART 1



For this Lab, we used the previous lab's (Lab 5) topology and configuration except for the routing. We had changed the routing configuration for this to OSPF (Open Shortest Path Finder).

Configuration

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router OSPF 10
R2(config-router)#network 10.6.1.8 0.0.0.7 area 0
R2(config-router)#network 10.6.1.200 0.0.0.7 area 0
R2(config-router)#network 10.6.1.200 0.0.0.7 area 0
*Mar  1 00:06:48.279: %OSPF-5-ADJCHG: Process 10, Nbr 10.6.1.206 on FastEthernet0/1 from LOADING to FULL, Loading Done
R2(config-router)#network 10.6.1.168 0.0.0.7 area 0
R2(config-router)#network 10.6.1.176 0.0.0.7 area 0
R2(config-router)#network 10.6.1.16 0.0.0.7 area 0
R2(config-router)#end
```

Configuration for the OSPF routing was done quite simply with two commands. First, mention the routing protocol that we were going to use (which is OSPF) followed by an ID (in this case 10).

Then the networks it was advertising followed by the area ID it was working under. Since we have only one network split into 5 or more subnets we will use the same area ID for all routers within the network.

The configuration was done for all the routers the same way with their respective networks.

The Neighbours

R1

```
R1#sh ip ospf neigh

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.6.1.197        1   FULL/DR         00:00:37    10.6.1.197      FastEthernet2/0
10.6.1.189        1   FULL/BDR        00:00:36    10.6.1.189      FastEthernet1/0
10.6.1.205        1   FULL/BDR        00:00:33    10.6.1.205      FastEthernet0/1
```

R2

```
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.6.1.189        1   FULL/BDR        00:00:36    10.6.1.181      FastEthernet2/0
10.6.1.197        1   FULL/BDR        00:00:30    10.6.1.173      FastEthernet1/0
10.6.1.206        1   FULL/DR         00:00:35    10.6.1.206      FastEthernet0/1
R2#
```

R3

```
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.6.1.205        1   FULL/DR         00:00:31    10.6.1.182      FastEthernet2/0
10.6.1.206        1   FULL/DR         00:00:32    10.6.1.190      FastEthernet1/0
10.6.1.197        1   FULL/BDR        00:00:31    10.6.1.166      FastEthernet0/1
R3#
```

R4

```
Neighbor ID     Pri   State           Dead Time   Address         Interface
10.6.1.206        1   FULL/BDR        00:00:31    10.6.1.198      FastEthernet2/0
10.6.1.189        1   FULL/DR         00:00:33    10.6.1.165      FastEthernet1/0
10.6.1.205        1   FULL/DR         00:00:37    10.6.1.174      FastEthernet0/1
R4#
```

We can see each of the routers within our network has a corresponding neighbor linked to them so that they can navigate their traffic to wherever they want within the network.

The neighbor ID is nothing but the router ID, it takes the highest IP by default to be its ID. Pri is the priority which in this case is 1 for all routers within the table. The state which is FULL/DR and FULL/BDR means it will route packets through the designated router and if that link were to fail it has a backup designated router to link the packets to. The dead time is the time it took for the routers to respond to the “Hello packet” sent by another router.

The address is the IP address the router's link is connected to and its corresponding interface.

Wireshark

We start a ping from PC 11 (10.6.1.25/29) to PC 7 (10.6.1.1/29) and everything seems to work fine until we shut down the interface 10.6.1.198 (R1) and we can see that even after the shutdown of the interface the packets took a small timeout and then takes a backup route to reach the destination it was intended to and the way this works is through something called LS update packets which we will see in a Wireshark trace.

```
R1(config)#int f2/0
R1(config-if)#shut
R1(config-if)#
```

```
84 bytes from 10.6.1.1 icmp_seq=45 ttl=62 time=60.010 ms
84 bytes from 10.6.1.1 icmp_seq=46 ttl=62 time=60.889 ms
10.6.1.1 icmp_seq=47 timeout
10.6.1.1 icmp_seq=48 timeout
10.6.1.1 icmp_seq=49 timeout
84 bytes from 10.6.1.1 icmp_seq=50 ttl=60 time=123.435 ms
84 bytes from 10.6.1.1 icmp_seq=51 ttl=60 time=104.458 ms
84 bytes from 10.6.1.1 icmp_seq=52 ttl=60 time=106.745 ms
84 bytes from 10.6.1.1 icmp_seq=53 ttl=60 time=105.860 ms
84 bytes from 10.6.1.1 icmp_seq=54 ttl=60 time=105.554 ms
84 bytes from 10.6.1.1 icmp_seq=55 ttl=60 time=105.341 ms
84 bytes from 10.6.1.1 icmp_seq=56 ttl=60 time=107.103 ms
84 bytes from 10.6.1.1 icmp_seq=57 ttl=60 time=105.121 ms
84 bytes from 10.6.1.1 icmp_seq=58 ttl=60 time=106.693 ms
84 bytes from 10.6.1.1 icmp_seq=59 ttl=60 time=106.077 ms
84 bytes from 10.6.1.1 icmp_seq=60 ttl=60 time=105.929 ms
84 bytes from 10.6.1.1 icmp_seq=61 ttl=60 time=105.570 ms
84 bytes from 10.6.1.1 icmp_seq=62 ttl=60 time=106.376 ms
84 bytes from 10.6.1.1 icmp_seq=63 ttl=60 time=106.214 ms
84 bytes from 10.6.1.1 icmp_seq=64 ttl=60 time=106.036 ms
84 bytes from 10.6.1.1 icmp_seq=65 ttl=60 time=106.044 ms
84 bytes from 10.6.1.1 icmp_seq=66 ttl=60 time=106.293 ms
84 bytes from 10.6.1.1 icmp_seq=67 ttl=60 time=105.513 ms
84 bytes from 10.6.1.1 icmp_seq=68 ttl=60 time=105.517 ms
84 bytes from 10.6.1.1 icmp_seq=69 ttl=60 time=105.335 ms
84 bytes from 10.6.1.1 icmp_seq=70 ttl=60 time=106.244 ms
84 bytes from 10.6.1.1 icmp_seq=71 ttl=60 time=105.900 ms
84 bytes from 10.6.1.1 icmp_seq=72 ttl=60 time=105.037 ms

PC11>
```

35	14.108736	10.6.1.25	10.6.1.1	ICMP	98	Echo (ping) request id=0xdd97, seq=43/11008, ttl=63 (reply in 36)
36	14.139292	10.6.1.1	10.6.1.25	ICMP	98	Echo (ping) reply id=0xdd97, seq=43/11008, ttl=63 (request in 35)
37	15.192086	10.6.1.25	10.6.1.1	ICMP	98	Echo (ping) request id=0xde97, seq=44/11264, ttl=63 (reply in 38)
38	15.221870	10.6.1.1	10.6.1.25	ICMP	98	Echo (ping) reply id=0xde97, seq=44/11264, ttl=63 (request in 37)
39	15.372657	10.6.1.198	224.0.0.5	OSPF	94	Hello Packet
40	16.276278	10.6.1.25	10.6.1.1	ICMP	98	Echo (ping) request id=0xdf97, seq=45/11520, ttl=63 (reply in 41)
41	16.305218	10.6.1.1	10.6.1.25	ICMP	98	Echo (ping) reply id=0xdf97, seq=45/11520, ttl=63 (request in 40)
42	17.360011	10.6.1.25	10.6.1.1	ICMP	98	Echo (ping) request id=0xe097, seq=46/11776, ttl=63 (reply in 43)
43	17.390018	10.6.1.1	10.6.1.25	ICMP	98	Echo (ping) reply id=0xe097, seq=46/11776, ttl=63 (request in 42)
44	17.736384	c4:01:40:78:00:20	CDP/VTP/DTP/PAgP/UD…	CDP	60	Device ID: R1
45	18.322241	10.6.1.197	224.0.0.5	OSPF	122	LS Update
46	18.444861	10.6.1.25	10.6.1.1	ICMP	98	Echo (ping) request id=0xe197, seq=47/12032, ttl=63 (no response found!)
47	20.464640	10.6.1.25	10.6.1.1	ICMP	98	Echo (ping) request id=0xe397, seq=48/12288, ttl=63 (no response found!)
48	20.840005	10.6.1.197	224.0.0.5	OSPF	94	Hello Packet
49	22.480795	10.6.1.25	10.6.1.1	ICMP	98	Echo (ping) request id=0xe597, seq=49/12544, ttl=63 (no response found!)
50	23.306604	10.6.1.197	10.6.1.198	OSPF	122	LS Update
51	23.592044	c4:04:34:90:00:20	c4:04:34:90:00:20	LOOP	60	Reply
52	27.861941	10.6.1.197	10.6.1.198	OSPF	122	LS Update

We can see there are three types of packets within this trace. The first is the ICMP protocol which is your usual PING request and reply. Second is your “Hello” packets which are sent by routers via a multicast address to constantly check if the routers are alive and responding. Third is an LS update packet which is only sent if there is a change in the OSPF table or if something happens to the router itself (i.e an interface shutdown or a router shutdown). These packets are sent to all the routers in the designated area to tell them that the interface is not working anymore and this route is impossible. After hearing this routers change their OSPF tables to the updated one and their DR to BDR if the corresponding router was the DR.

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.6.1.189	1	FULL/DR	00:00:39	10.6.1.165	FastEthernet1/0
10.6.1.205	1	FULL/DR	00:00:35	10.6.1.174	FastEthernet0/1

We can see the router table change here in R2.

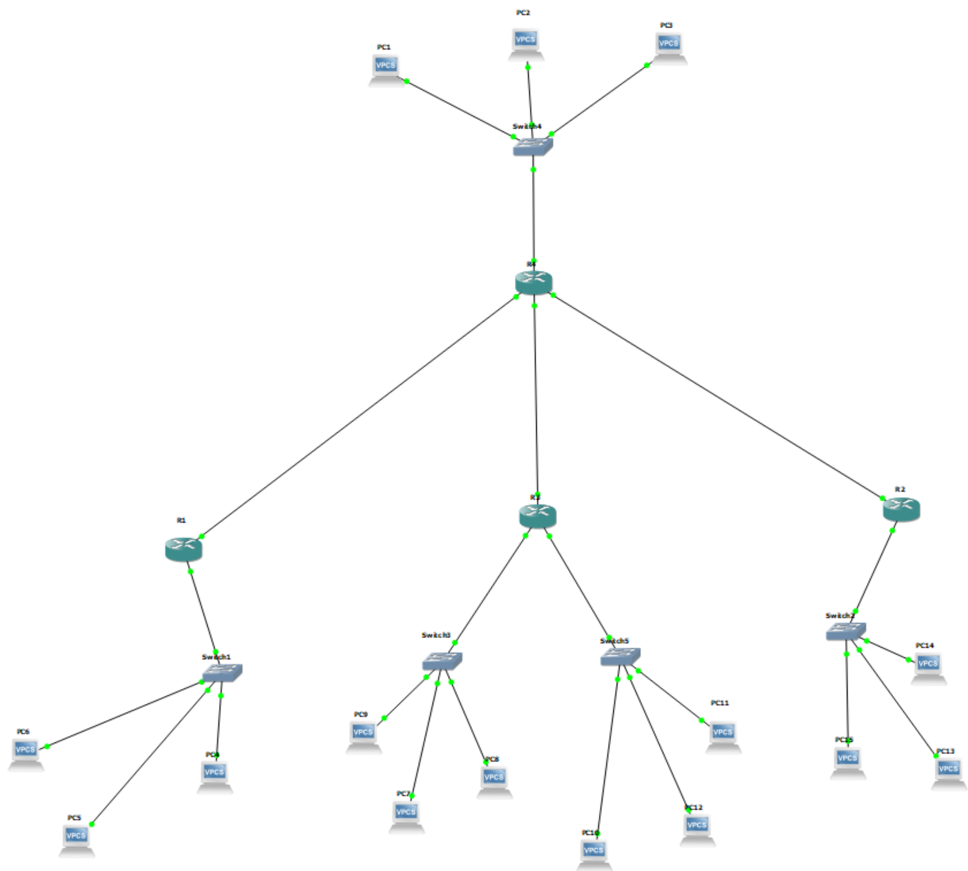
1	0.000000	10.6.1.197	224.0.0.5	OSPF	94 Hello Packet
2	6.261804	c4:04:34:90:00:20	c4:04:34:90:00:20	LOOP	60 Reply
3	8.729939	10.6.1.197	224.0.0.5	OSPF	146 LS Update
4	9.952258	10.6.1.197	224.0.0.5	OSPF	94 Hello Packet
5	16.253819	c4:04:34:90:00:20	c4:04:34:90:00:20	LOOP	60 Reply
6	19.839021	10.6.1.197	224.0.0.5	OSPF	90 Hello Packet

The same goes for when a router were to shut down instantly. An update packet is sent to all the routers telling them to update the routes/tables.

38	114.329543	10.6.1.198	10.6.1.197	OSPF	78 DB Description
39	114.344546	10.6.1.197	10.6.1.198	OSPF	78 DB Description
40	114.344546	10.6.1.197	10.6.1.198	OSPF	258 DB Description
41	114.360264	10.6.1.198	10.6.1.197	OSPF	98 DB Description
42	114.375159	10.6.1.197	10.6.1.198	OSPF	78 DB Description
43	114.390799	10.6.1.198	10.6.1.197	OSPF	78 DB Description
44	114.390799	10.6.1.198	10.6.1.197	OSPF	166 LS Request
45	114.405803	10.6.1.197	10.6.1.198	OSPF	78 DB Description
46	114.405803	10.6.1.197	10.6.1.198	OSPF	522 LS Update

Once the router comes back up it sends a request packet asking for the description of the neighboring routing tables (DB Description) so that it can update its own routing table and others' routing tables via an LS update packet.

PART 2



For this Lab, we used the previous lab's (Lab 5) topology and configuration except for the routing. We changed the routing configuration for this to EIGRP (Open Shortest Path Finder).

The full extent of EIGRP was not explored because of the limitation of the topology. However, the main points are covered.

Configuration

```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router EIGRP 10
R1(config-router)#network 192.168.0.0 0.0.0.7
R1(config-router)#network 192.168.0.200 0.0.0.7
R1(config-router)#no auto-summary
R1(config-router)#end
```

Configuration for the EIGRP routing was done quite simply also mentioning the EIGRP ID and then the network along with the wildcard mask.

The Neighbours

R1

```
IP-EIGRP neighbors for process 10
H   Address                 Interface        Hold Uptime    SRTT   RTO  Q   Seq
                                             (sec)          (ms)        Cnt Num
0   192.168.0.205           Fa0/1              10 00:02:36  645   3870  0   22
```

R2

```
IP-EIGRP neighbors for process 10
H   Address                 Interface        Hold Uptime    SRTT   RTO  Q   Seq
                                             (sec)          (ms)        Cnt Num
0   192.168.0.197           Fa0/1              13 00:03:03 1012   5000  0   23
R3#
```

R3

```
IP-EIGRP neighbors for process 10
H   Address                 Interface        Hold Uptime    SRTT   RTO  Q   Seq
                                             (sec)          (ms)        Cnt Num
0   192.168.0.189           Fa1/0              11 00:03:07   78    468  0   24
R2#
```

R4

```
IP-EIGRP neighbors for process 10
H   Address                 Interface        Hold Uptime   SRTT   RTO  Q  Seq
                                             (sec)         (ms)       Cnt Num
2   192.168.0.190           Fa1/0              9 00:03:55   85    510  0  6
1   192.168.0.198           Fa0/1             11 00:04:09  100    600  0  8
0   192.168.0.206           Fa0/0             11 00:04:18   96    576  0  9
R4#
```

We can see each of the routers within our network has a corresponding neighbor linked to them so that they can navigate their traffic to wherever they want within the network.

The interface each of the networks corresponds to with some additional information such as the hold time which tracks the time remaining when a router is deemed unreachable (Due to hello packets not being acknowledged). Then there is the uptime which shows how long the neighbour relationship is in real time.

The SRTT measures the estimated time from the router to the neighbor to send a packet and back. Then the RTO which is how long the router would wait before re-transmitting the packet.

Wireshark

```
PC6> ping 192.168.0.25 -t
192.168.0.25 icmp_seq=1 timeout
192.168.0.25 icmp_seq=2 timeout
84 bytes from 192.168.0.25 icmp_seq=3 ttl=61 time=90.161 ms
84 bytes from 192.168.0.25 icmp_seq=4 ttl=61 time=89.913 ms
84 bytes from 192.168.0.25 icmp_seq=5 ttl=61 time=90.701 ms
84 bytes from 192.168.0.25 icmp_seq=6 ttl=61 time=89.567 ms
84 bytes from 192.168.0.25 icmp_seq=7 ttl=61 time=91.397 ms
84 bytes from 192.168.0.25 icmp_seq=8 ttl=61 time=89.612 ms
84 bytes from 192.168.0.25 icmp_seq=9 ttl=61 time=89.980 ms
84 bytes from 192.168.0.25 icmp_seq=10 ttl=61 time=91.990 ms
84 bytes from 192.168.0.25 icmp_seq=11 ttl=61 time=90.201 ms
```

We can see the ping from PC 6 (192.168.0.1) to PC 14 (192.168.0.25) seems to work. However, we are unable to test if an interface were to go down simply because the topology is a single-point failure topology and we have no other connections anywhere else. But for EIGRP it should change route automatically if the primary route were to fail for some reason

1	0.000000	192.168.0.205	224.0.0.10	EIGRP	74 Hello
2	3.564561	192.168.0.206	224.0.0.10	EIGRP	74 Hello
3	4.437832	c4:04:4b:54:00:00	c4:04:4b:54:00:00	LOOP	60 Reply
4	5.399640	192.168.0.205	224.0.0.10	EIGRP	74 Hello

We can see that for EIGRP the way it detects and tells if its neighbors are still up and active is through "Hello" packets. 192.168.0.205 sends a "Hello" packet and 192.168.0.206 sends one back to say to the router that it is still here through multicast.

20	15.310247	192.168.0.197	192.168.0.198	EIGRP	60 Update
21	15.325257	192.168.0.198	192.168.0.197	EIGRP	112 Update
22	15.340019	192.168.0.198	192.168.0.197	EIGRP	60 Hello (Ack)
23	15.354873	192.168.0.197	192.168.0.198	EIGRP	199 Update
24	15.369733	192.168.0.198	192.168.0.197	EIGRP	112 Update
25	15.384623	192.168.0.197	192.168.0.198	EIGRP	60 Hello (Ack)
26	15.384623	192.168.0.198	192.168.0.197	EIGRP	60 Hello (Ack)
27	15.399511	192.168.0.197	224.0.0.10	EIGRP	112 Update
28	15.414408	192.168.0.198	192.168.0.197	EIGRP	60 Hello (Ack)
29	15.429415	192.168.0.197	192.168.0.198	EIGRP	199 Update
30	15.444518	192.168.0.198	192.168.0.197	EIGRP	60 Hello (Ack)
31	17.900962	192.168.0.197	224.0.0.10	EIGRP	74 Hello
32	18.187518	192.168.0.198	224.0.0.10	EIGRP	74 Hello

For this one, I had shut one interface down and saw how EIGRP would tell other routers that this link is no longer valid, and the way it tells it is through Update packets similar to OSPF. After it sends those update packets it then sends an ACK or acknowledgment packet to confirm it has received the update packet. Sometimes it also sends a query to ask the router if there are any successor routes (alternative routes) to the missing path.

76	104.738777	192.168.0.197	224.0.0.10	EIGRP	74 Hello
77	110.084075	192.168.0.197	224.0.0.10	EIGRP	74 Hello
78	115.457210	192.168.0.197	224.0.0.10	EIGRP	74 Hello

For this one, we had shut down a router fully to see how the neighboring router would react. We can see that 192.168.0.197 keeps sending "Hello" packets but doesn't get a response back from the neighboring router and hence will remove it from the routing table below.

```
IP-EIGRP neighbors for process 10
H   Address                 Interface       Hold Uptime   SRTT   RTO  Q  Seq
                                            (sec)         (ms)       Cnt Num
2   192.168.0.190           Fa1/0             10 00:07:08   79   474  0   31
0   192.168.0.206           Fa0/0             14 00:07:08   82   492  0   33
R4#
```

Over we can see that R4 has removed R3 from its neighbor list due to it being unresponsive.